Journal of Computational Physics 230 (2011) 1454–1464

Contents lists available at ScienceDirect

Journal of Computational Physics

journal homepage: www.elsevier.com/locate/jcp

Neutral gas free molecular flow algorithm including ionization and walls for use in plasma simulations

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A R T I C L E I N F O

Article history:
Received 3 May 2010
Received in revised form 16 September 2010
Accepted 8 November 2010
Available online 3 December 2010

Keywords:
Plasma simulation
Partially ionized gases

A B S T R A C T

Several plasma devices, including Hall and ion thrusters, operate by ionizing a low density neutral gas for which the mean free path between collisions of gas molecules is greater than typical device dimensions. In general, the discrete-particle algorithms used to calculate the neutral gas ignore velocity changes due to collisions between gas molecules. However, particle algorithms are a source of unphysical statistical noise that may detract from the study of the plasma physics, the prime purpose of most simulations. In this paper we present a new neutral gas algorithm for use in plasma simulation codes that exploits the fact that very few collisions change the velocity of neutral gas molecules. The algorithm assumes that the particle velocity distribution function for neutrals emitted from a given surface remains unchanged except for a scale factor that reflects the loss of neutrals to ionization. The sources of neutrals may be gas inlets, and isotropic, thermally accommodated, gas molecules coming off chamber surfaces including recombined ions. The algorithm is implemented in two dimensions (R–Z) with emitting surfaces represented as surfaces of revolution. The advantage of this algorithm over the conventional particle approach is the absence of statistical noise.

1. Introduction

Several plasma devices, including Hall and ion thrusters operate by ionizing a low density neutral gas. In many cases the physics of the ionization and the electron and ion dynamics are quite complex, and controlled by time dependent electric and magnetic fields. In these devices the neutral gas Knudsen number is usually much greater than one. In this regime of "free molecular flow" continuum fluid models are not applicable. While collisions between neutrals are not very important, ionizing collisions with electrons are of utmost importance as is reemission from walls of both gas molecules and recombined ions. The long mean free path for collisions between gas molecules is well recognized by the developers of plasma simulations codes, and the particle algorithms used to calculate neutral gas in general ignore velocity changes due to such collisions. Computer models used to simulate Hall and ion thrusters typically simulate the neutral gas using particle in cell (PIC) algorithms combined with direct simulation Monte Carlo (DSMC) to account for ionization [1,2].

The PIC algorithm can lead to artificially large density fluctuations due to particle statistics, [3] a source of unphysical statistical noise that detracts from the study of the plasma physics, the prime purpose of most simulation. For example, the time dependent neutral gas density computed along the middle of the channel in a 6 kW laboratory Hall thruster [4] using the hybrid-PIC simulation code "HPHall" [1,2], is shown in Fig. 1. The parameters for this calculation were not

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0021-9991/$ - see front matter
doi:10.1016/j.jcp.2010.11.013

Nomenclature

b	boolean ray block function ($b = 1$ ray unblocked, $b = 0$ ray blocked)
$\bar{c}$	mean molecular speed
$f(\mathbf{x}, \mathbf{v}, t)$	molecular velocity distribution function
$\mathrm{f}(\mathbf{x}, \mathbf{v})$	normalized velocity distribution function
i	cell index
j	emitting surface index
J	collective emitting surface index
k	cell edge index
m	triangular cell index
M	molecular mass
n	neutral particle density
T	emitted gas temperature
$\mathbf{u}$	cell edge normal vector
ν	ionization frequency
Γ	neutral particle flux
Ω	solid angle

optimized to minimize statistical noise, nor were they intentionally modified to increase the noise. The statistical noise can be reduced by increasing the number of macro particles, but additional particles increase computation time. Careful tailoring of the weighting and emission algorithms also can reduce the PIC statistical noise [2].

While the PIC algorithm is very popular for modeling collisionless plasmas, other techniques are commonly employed for modeling free molecular flow. Free molecular flow in the presence of walls has been traditionally modeled [5] using the same algorithms used for diffuse photon transport in thermal calculations [6]. In both cases, the incident molecules or photons are assumed to be fully accommodated on surfaces they hit, and the fraction remained is assumed to be a Lambertian (cosine) distribution. The molecular flux incident on a given surface depends on the view factor between that surface and surfaces emitting molecules. Most spacecraft contamination codes also use an algorithm that is based upon this approach [7].

The view factor approach assumes that the transit time between surfaces is negligible compared with other timescales in the problem. However, in most plasma devices, the neutral gas molecules move slower than ions and electrons so preserving the time dependence of the neutral density is essential. The algorithm presented below uses view factors to generate approximate molecular velocity distribution functions on a grid throughout the thruster volume, and uses a mass-conserving first-order upwind algorithm to time step the neutral gas between grid cells. As a result, although the neutral velocities are fixed, the neutral density is time dependent. The goal of the algorithm is to provide a solution for the neutral gas density that does

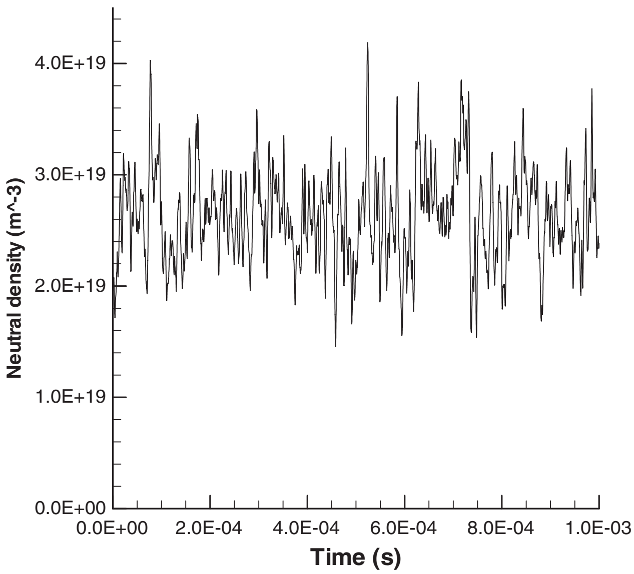

Fig. 1. Mid-channel neutral gas density for a 6 kW Hall thruster calculated using HPHall.

not introduce numerical errors into the plasma simulation from discrete-particle statistics; it is not intended as a tool to study the nuances of free-molecular gas dynamics.

2. The Algorithm

The fundamental concept of the algorithm is to solve for the neutral gas density by integrating forward in time the linear Boltzmann equation in the absence of forces. The equation for the neutral gas including ionization but ignoring all other collisions is

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f = -\nu f, \tag{1}$$

where

$$f \equiv f(\mathbf{x}, \mathbf{v}, t),$$
$$n \equiv n(\mathbf{x}, t) = \int f d\mathbf{v},$$
$$\nu \equiv \nu^{\text{ionize}}(\mathbf{x}, t).$$

The right-hand term in Eq. (1) represents the rate at which gas molecules are removed from the distribution function due to ionization. It is noted that velocity changes due to charge exchange collisions have been ignored.

The discretization of the velocity distribution function is the key step in developing a finite-difference algorithm that takes advantage of the straight line molecular trajectories. A velocity vector can be expressed in terms of the speed, s, and a normalized direction vector, $\mathbf{\Omega}$.

$$\begin{aligned}
\mathbf{v} &= (s\cos\phi\sin\theta, s\sin\phi\sin\theta, s\cos\theta), \\
\mathbf{\Omega} &\equiv (\sin\phi\sin\theta, \cos\phi\sin\theta, \cos\theta), \\
d\mathbf{\Omega} &\equiv \sin\theta d\theta d\phi, \\
n(\mathbf{x}, t) &= \iint f(\mathbf{x}, s, \mathbf{\Omega}, t) ds d\mathbf{\Omega}.
\end{aligned} \tag{2}$$

Consider a discretized physical domain consisting of computational elements (or cells), each identified by index "i". The gas density at $\mathbf{x}_i$, the position vector of the center of cell i, can be approximated as a sum over discrete solid angles, $\delta\mathbf{\Omega}_{i,j}$, and speed intervals, ℓ,

$$\begin{aligned}
n_i^t &\equiv n(\mathbf{x}_i, t) = \sum_{j,\ell} n_{i,j,\ell}^t, \\
n_{i,j,\ell}^t &\equiv \int_{\delta\Omega_{i,j}} \int_{s_{\ell-1}}^{s_\ell} f(\mathbf{x}_i, s, \mathbf{\Omega}, t) ds d\mathbf{\Omega},
\end{aligned} \tag{3}$$

where $n_{i,j,\ell}^t$ is the contribution to the density in cell i at time t from gas molecules emanating from surface j with speeds s where $s_{\ell-1} \leqslant s < s_\ell$.

It is further assumed that the speed can be factored out of each discrete angle distribution function, and within each discrete solid angle, there is no angular dependence:

$$f(\mathbf{x}_i, s, \mathbf{\Omega}, t) \Rightarrow \bar{f}_{i,j}^t g_j(s), \tag{4}$$

where

$$\int_0^\infty g_j(s) ds = 1. \tag{5}$$

A single speed interval is used in the implementation below, and the speed integral is over the interval $(0, \infty)$. As a result, the cell-centered density can be written as:

$$\begin{aligned}
n_i^t &= \sum_j n_{i,j}^t, \\
n_{i,j}^t &\equiv \int_{\delta\Omega_{i,j}} \bar{f}_{i,j}^t \delta\Omega.
\end{aligned} \tag{6}$$

The errors introduced by using a single speed interval are discussed later in this paper.

In this algorithm the discrete solid angles $\delta\Omega_{i,j}$ are chosen such that at the center of every cell i, the solid angle subtended by $\delta\Omega_{i,j}$ corresponds to the same gas-emitting surface area, j, on the boundary of the computational mesh. This method is analogous to those using view factors to calculate the surface-to-surface transport of photons in thermal models or the transport of molecules in spacecraft contamination codes. The view factor approach is applied to a time dependent solution of the Boltzmann equation inside each cell volume. Fig. 2 illustrates how $\delta\Omega_{i,j}$ subtends a smaller solid angle the farther the cell, i, is from the emitting surface, j. Fig. 2 is a simplification of the actual 2-D axisymmetric algorithm because the emitting surfaces in the 2-D axisymmetric algorithm are surfaces of revolution and the solid angles form stripes around the unit sphere. Since

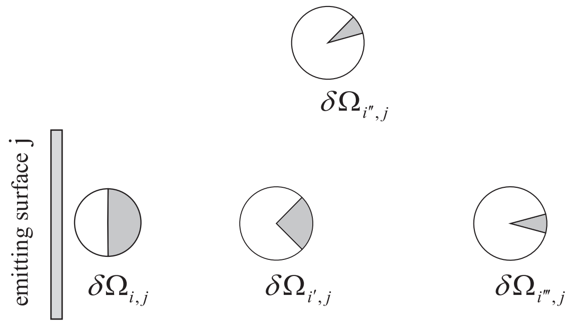

Fig. 2. Discrete solid angle, $\delta\Omega_{i,j}$, in a cell, i, depends on its location with respect to the gas emitting surface, j.

the gas molecules all move in straight lines, in any cell i, only molecules emitted from surface j contribute to the distribution function over the solid angle $\delta\Omega_{i,j}$.

Integration of Eq. (1) over speed for a given discrete solid angle,

$$\frac{\partial}{\partial t}\int_{\delta\Omega_{i,j}}\int_0^\infty f(\mathbf{x}_i,s,\mathbf{\Omega},t)dsd\Omega+\int_{\delta\Omega_{i,j}}\int_0^\infty \mathbf{v}\cdot\nabla f(\mathbf{x}_i,s,\mathbf{\Omega},t)dsd\Omega=-v\int_{\delta\Omega_{i,j}}\int_0^\infty f(\mathbf{x}_i,s,\mathbf{\Omega},t)dsd\Omega \tag{7}$$

results in analogues to the fluid mass continuity equation for each emitting surface, j.

$$\frac{\partial}{\partial t}n_{i,j}^t+\int_{\delta\Omega_i}\int_0^\infty \mathbf{v}\cdot\nabla f(\mathbf{x}_i,s,\mathbf{\Omega},t)dsd\Omega=-v_i^t n_{i,j}^t. \tag{8}$$

The computational domain is divided into 2-D convex polygon cells, i, of volume, V_i. The cell volume is bounded by surfaces, k, each with surface area A^k, as shown in Fig. 3.

The continuity equations, Eq. (8), are solved conservatively using Gauss's theorem to evaluate the divergence in terms of fluxes across cell surfaces. Each surface, k, separates two cells, i and i'. The upwind, flux flowing out of cell i, into cell i' for molecules starting from surface j is found by integrating over each surface k as follows:

$$\Gamma_{i,j}^k\equiv\frac{n_{i,j}^t\bar{c}_j}{A^k\delta\Omega_j}\int_{A^k}\int_{\delta\Omega_j}\max(0,\mathbf{\Omega}\bullet\mathbf{u}_{i\to i'}^k)d\Omega dA, \tag{9}$$

where $\mathbf{u}_{i\to i'}^k$ is the unit vector normal to surface A^k pointing outward from cell i into cell i', and $\bar{c}_j$ is the mean molecular speed associated with surface j,

$$\bar{c}_j\equiv\int_0^\infty g_j(s)sds. \tag{10}$$

Rewriting in terms of an average one-sided velocity across the surface gives

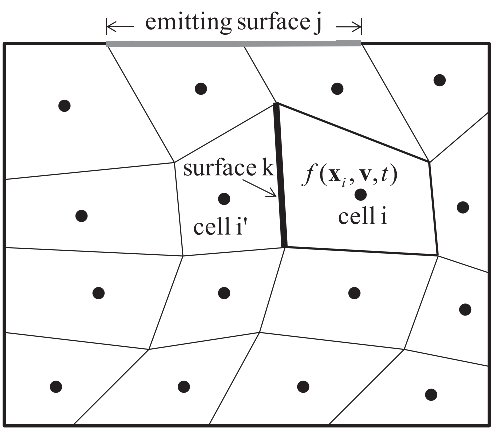

Fig. 3. Example computational region, divided into cells, i, bounded by surfaces, k. Neutral gas enters the computational region from boundary surfaces, j, that may be composed of several boundary cell edges.

$$\begin{aligned}\Gamma_{i,j}^{k} &= n_{i,j}^{t}\bar{v}_{i,j}^{k},\\ \bar{v}_{i,j}^{k} &\equiv \frac{\bar{c}_j}{A^k\delta\Omega_j}\int_{A^k}\int_{\delta\Omega_j}\max(0,\mathbf{\Omega}\bullet\mathbf{u}_{i\to i\prime}^{k})d\Omega dA.\end{aligned} \tag{11}$$

Eq. (8) may be written in finite difference form as:

$$\frac{\left(n_{i,j}^{t+1}-n_{i,j}^{t}\right)}{\Delta t}V_i = \sum_k A^k\left(\Gamma_{i',j}^{k}-\Gamma_{i,j}^{k}\right) - v_i^t n_{i,j}^t V_i. \tag{12}$$

Replacing the fluxes with the product of the upwind densities and surface centered velocities results in the following equation, Eq. (13), to advance in time the contribution to the gas density in cell i from emitting surface j.

$$n_{i,j}^{t+1} = n_{i,j}^{t} + \Delta t\left(\sum_k \frac{A^k}{V_i}\left(n_{i',j}^{t}v_{i',j}^{k} - n_{i,j}^{t}v_{i,j}^{k}\right) - v_i^t n_{i,j}^{t}\right). \tag{13}$$

The main approximation in the present algorithm is to assume that in the equation above the average velocities are independent of time. They are calculated only once, assuming no ionization using the velocity distribution of the source. Fig. 4 shows how across a single surface there can be two effective velocities associated with the gas from a single emitting surface, j. The first, $v_{i,j}^{k}$, is the velocity at which gas molecules leave cell i across surface k. The second, $v_{i',j}^{k}$, is the velocity at which particles enter cell i from an adjacent cell, i'.

Note that while Eq. (13) is derived from the Eq. (1), the distribution function no longer appears explicitly. Only partial densities and velocity-averaged inner products with the surface normals appear. This equation has the form of the fluid continuity equation with one major difference: there can be two velocities associated with each edge; the first is for gas entering the cell, the second for gas leaving the cell.

The determination of these velocities on a computational mesh with cylindrical geometry, Fig. 5, proceeds as follows. First, a surface of revolution is generated for each emitting surface j. The surface is then divided into triangles. An example of this for a radial surface that includes the axis is shown in Fig. 6. The identical algorithm applies to sloped (conical) and horizontal (cylindrical) surfaces. Note that by symmetry, only half a circle is needed. Fig. 6 shows uniform angular spacing but the angular spacing may be varied so that the smallest triangles are nearest the R–Z plane. This would provide higher resolution for points closest to the surface.

For each triangle, a ray is drawn from the center of cell center i to the center of the triangle. If the ray intersects any boundary surface prior to reaching the cell center the ray is considered blocked. That is, with $r_{i,j}$ being the ray from the center of cell i to the center of triangle m, then the boolean ray block function is

$$b_{i,m} = \begin{cases}1, & r_{i,m}\text{unblocked},\\ 0, & r_{i,m}\text{blocked}.\end{cases} \tag{12}$$

The discrete solid angles $\delta\Omega_{i,j}$ are found by summing the view factors of the triangles with unblocked rays.

$$\delta\Omega_{i,j} = \sum_{m\in j} b_{i,m}\delta\Omega_{i,m}. \tag{13}$$

The view factor for each triangle is calculated using the algorithm of Oosterom and Strackee [8].

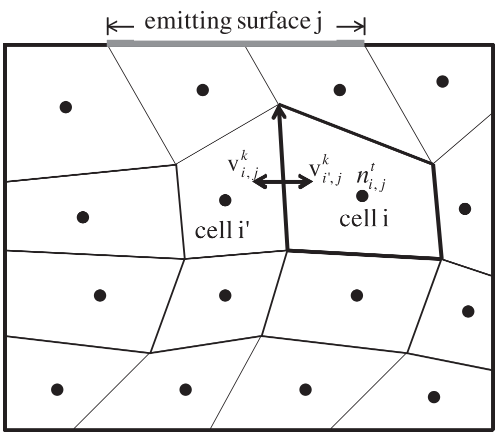

Fig. 4. The contribution from each emitting surface j to the cell-centered gas density is found by integrating the mass continuity equation using the upwind density and time independent velocities, which allow gas to both enter and leave the cell, i, across a surface, k.

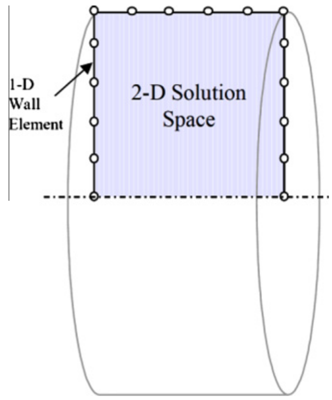

Fig. 5. Edges on a 2-D grid represent a surface of revolution.

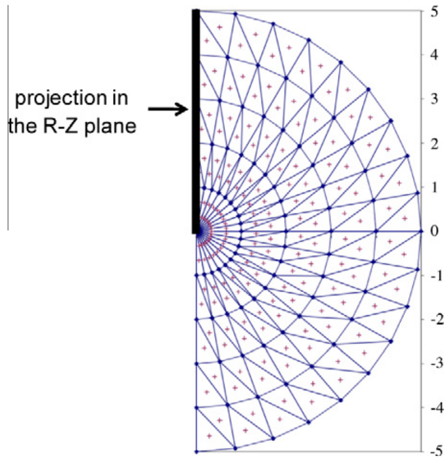

Fig. 6. Triangulated surface of revolution and its projection in the R–Z plane.

The flux across edge k leaving cell i is given by

$$\Gamma_{i,j}^{k} = \frac{\bar{c}_j}{4\pi} \sum_{m \in J} b_{m,i} \Omega_{m,i} \frac{\max(0, \mathbf{r}_{i,m} \bullet \mathbf{u}_{i \to i'}^{k})}{|\mathbf{r}_{i,m}|}, \tag{14}$$

where the average molecular speed, $\bar{c}_j$, is defined as:

$$\bar{c}_j \equiv \sqrt{\frac{8 k_B T_j}{\pi M}} \tag{15}$$

and T_j is the temperature of surface j.

The fluxes for an edge are calculated, as the average of fluxes at its two vertices as field points. In practice this has ensured that fluxes entering a cell also leave the cell, since each vertex flux contributes to two edges, frequently with opposite signs, as flux entering the cell along one edge and as flux leaving the cell along the other edge.

$$\langle \Gamma_{i,j}^{k} \rangle = \frac{1}{2} \left(\Gamma_{i,j}^{k}|_{\text{vertex1}} + \Gamma_{i,j}^{k}|_{\text{vertex2}} \right). \tag{14}$$

The velocity vector associated with the edge for gas that was emitted from surface j, and leaves cell i through edge k, $\mathrm{v}_{i,j}^{k}$, is calculated using the edge-centered fluxes and the cell-centered normalized density. Since the fluxes and the density are calculated at different locations, the velocities are limited to be less than or equal to the thermal speed

$$\mathrm{v}_{i,j}^{k} = \min \left(\frac{\langle \Gamma_{i,j}^{k} \rangle}{n_{i,j}}, \bar{c}_j \right). \tag{16}$$

These edge centered velocities are calculated only once for a given geometry and are assumed independent of time.

3. Results

The algorithm presented above has been implemented in a 2-D, R–Z, Hall effect thruster code, "Hall 2De" [9]. Hall 2De solves the governing laws for the partially-ionized gas generated by a Hall thruster on a 2-D axisymmetric computational mesh that is closely aligned with the applied magnetic field. Several numerical tests were performed to verify the accuracy of the view factors in the presence of shadowing. Another test was a direct comparison of the solutions produced by the (hybrid-PIC) code HPHall and Hall 2De. The comparison of the neutral gas density along the center of the thruster channel for an Aerojet BPT-4000 4.5 kW Hall thruster simulation is shown in Fig. 7. It is noted that the HPHall solution is averaged over thousands of cycles while the solution from the present algorithm is instantaneous. The two results are in excellent agreement until ionization has led to a loss of over ninety percent of the gas. About a centimeter past the channel exit plane, the gas density in both calculations is less than 1/1000 of the density near the anode.

Another direct comparison was made between the two codes for calculations of a 6 kW laboratory Hall thruster. Neutral gas density contour plots in the acceleration channel of the Hall thruster are shown in Fig. 8. The Hall 2De field-aligned mesh is coarser than the near-rectilinear mesh employed by HPHall. The gas densities are similar, with much of the differences attributable to the differences between the two codes in their respective plasma solutions.

Fig. 9 compares time dependent densities obtained by the two codes at the cells outlined in black in Fig. 8. At these locations ionization has reduced the neutral gas density by about half. Since the codes compute somewhat different plasma

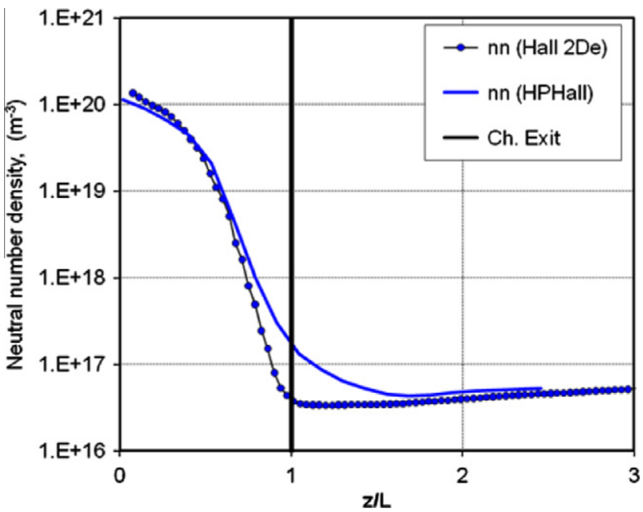

Fig. 7. Comparison between the neutral gas densities at the center of the thruster channel of a BPT-4000 as calculated by the 2-D (axisymmetric) codes HPHall and Hall 2De.

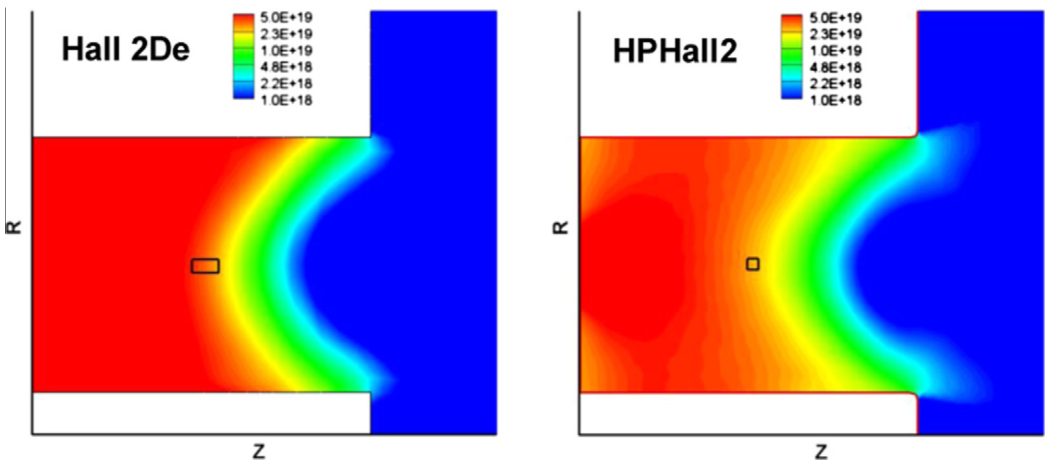

Fig. 8. Channel and near plume neutral gas densities ($\#/m^3$) calculated by the new algorithm in Hall 2De compared with time-averaged results of HPHall.

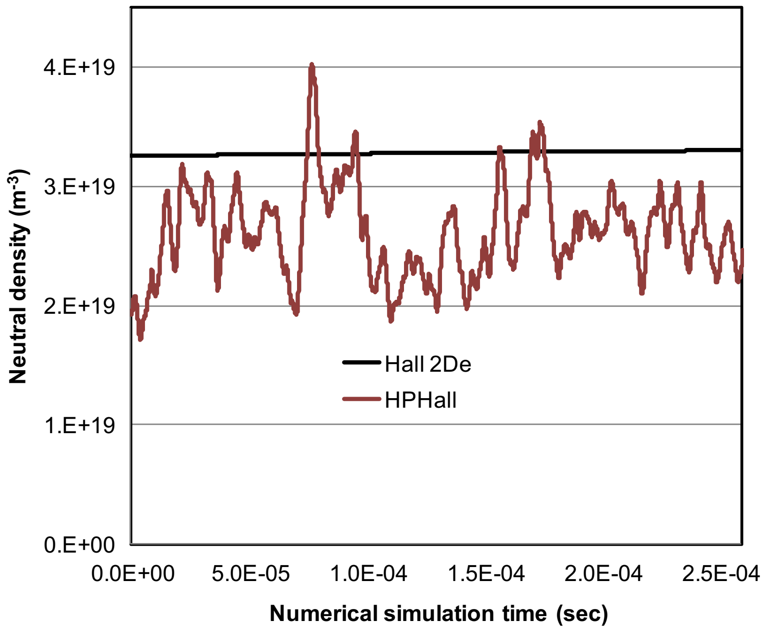

Fig. 9. Time dependent mid-channel neutral gas densities calculated by the new neutral gas algorithm in Hall 2De compared with PIC–Monte Carlo calculations from HPHall.

densities, the average values of the gas utilized at the particular locations are different. However, by contrast to the PIC–Monte Carlo calculations of HPHall, it is clear that the new algorithm eliminates numerical fluctuations.

No direct comparison was made of the relative speed of the two algorithms. However, after the velocities have been calculated initially, for each ensuing time step the number of operations to calculate the contribution to the density from an emitting surface, j, to a cell, i, using the new algorithm is the same order of magnitude as advancing a single particle in a PIC code and calculating its contribution to cell densities. For the calculation shown in Figs. 8 and 9, the boundary was divided into 10 emitting surfaces. Thus, the calculation took roughly the same computer time as a PIC code would take if it had an average of 10 macro particles per zone.

4. Discussion

The accuracy of the PIC algorithm is well established. This section presents an analysis of some approximations affecting the new algorithm's accuracy. The principal assumption in the algorithm is that the neutral gas molecules move in straight lines at constant velocity. In the Hall effect thruster simulations shown above, the peak neutral density was less than 10^{20}#/m^3. Assuming a molecular radius of 2.18 Å, the neutral–neutral collision mean free path is more than twice the channel width. This high density region is located upstream in the channel; everywhere else the mean free path is much longer. As long as the molecular mean free path is long compared with the channel dimensions, the straight line trajectory assumption is valid.

The second assumption is that the density is constant for each discrete solid angle. While related to the absence of momentum changing collisions, to be strictly valid, this assumption requires that the attenuation of density due to ionization be uniform across the discrete solid angles. However, the finite solid angle encompasses molecules with a spread in path lengths from emitting surfaces. Just as the accuracy of PIC improves with the number of particles, the accuracy of the algorithm above can be improved by increasing the number of the discrete solid angles, j. In the calculation shown in Fig. 8, there were only three solid angles to describe the surfaces in the channel (back wall, inner and outer walls). The anode and each small wall element individually emit gas, but there are only three discrete solid angles, each corresponding to one of the three emitting surfaces. Increasing the number of solid angles and decreasing the size of each surface j, would decreases the spread in each solid angle, thereby reducing errors due to path length differences.

Finally, the algorithm assumes that the speed distribution does not change. For molecules starting out as a Maxwellian, ionization changes the speed distribution function. The ionization collision loss term in Eq. (1) is independent of velocity. However, the transit time depends inversely on the speed of a particle; slower particles take more time to move a given distance and are more likely to be ionized, faster particles are less likely to be ionized. As a result, the speed distribution function changes with the degree of ionization. This effect can be quantified by examining this assumption for a 1-D model problem. In the model problem the density as a function of position, x, is given by

$$n(x) = \frac{2}{\sqrt{\pi}} \int e^{-\mathbf{v}^2} e^{-\alpha t(\mathbf{v})} d\mathbf{v}, \tag{17}$$

where α is the ionization rate.

Eliminating time from Eq. (17) results in an expression for the density as a function of position,

$$n(x) = \frac{2}{\sqrt{\pi}} \int e^{-\mathbf{v}^2} e^{-\alpha \frac{x}{\mathbf{v}}} d\mathbf{v} \tag{18}$$

using $t = x/v$. Assuming constant speed, the equivalent approximation to Eq. (5) results in a simple exponential decay of density, n_{cs}, with position:

$$n_{\mathrm{cs}}(x) = e^{-\alpha \frac{x}{\bar{\mathbf{v}}}}, \tag{19}$$

where

$$\bar{\mathbf{v}} \equiv \frac{\int \mathbf{v} e^{-\mathbf{v}^2} d\mathbf{v}}{\int e^{-\mathbf{v}^2} d\mathbf{v}}. \tag{20}$$

This choice for $\bar{\mathbf{v}}$ yields the correct neutral density in the absence of ionization.

Density as a function of position for both the exact solution and for the constant-speed approximation are shown in Fig. 10. In the exact solution the first molecules to be ionized are the ones that move slower. The constant-speed assumption ignores this velocity dependence and initially predicts that more gas remains. When over 90% of the gas has been ionized the remaining Maxwellian gas molecules have high velocities and move very quickly through the ionizing region. Thus, when most of the gas has been ionized, the constant-speed assumption predicts a higher degree of ionization than the exact solution.

A simple improvement to the algorithm is to have multiple velocity bins. The solution for the same model problem with two velocity bins is given in Eq. (21).

$$\begin{aligned}
n_{2\mathrm{speed}}(x) &= \frac{1}{2}\left(e^{-\alpha \frac{x}{\bar{\mathbf{v}}_1}} + e^{-\alpha \frac{x}{\bar{\mathbf{v}}_2}}\right), \\
\int_0^{\mathbf{v}_1} e^{-\mathbf{v}^2} d\mathbf{v} &= \frac{1}{2}\int_0^{\infty} e^{-\mathbf{v}^2} d\mathbf{v}, \\
\bar{\mathbf{v}}_1 &\equiv \frac{\int_0^{\mathbf{v}_1} \mathbf{v} e^{-\mathbf{v}^2} d\mathbf{v}}{\int_0^{\mathbf{v}_1} e^{-\mathbf{v}^2} d\mathbf{v}}, \\
\bar{\mathbf{v}}_2 &\equiv \frac{\int_{\mathbf{v}_1}^{\infty} \mathbf{v} e^{-\mathbf{v}^2} d\mathbf{v}}{\int_{\mathbf{v}_1}^{\infty} \mathbf{v} e^{-\mathbf{v}^2} d\mathbf{v}}.
\end{aligned} \tag{21}$$

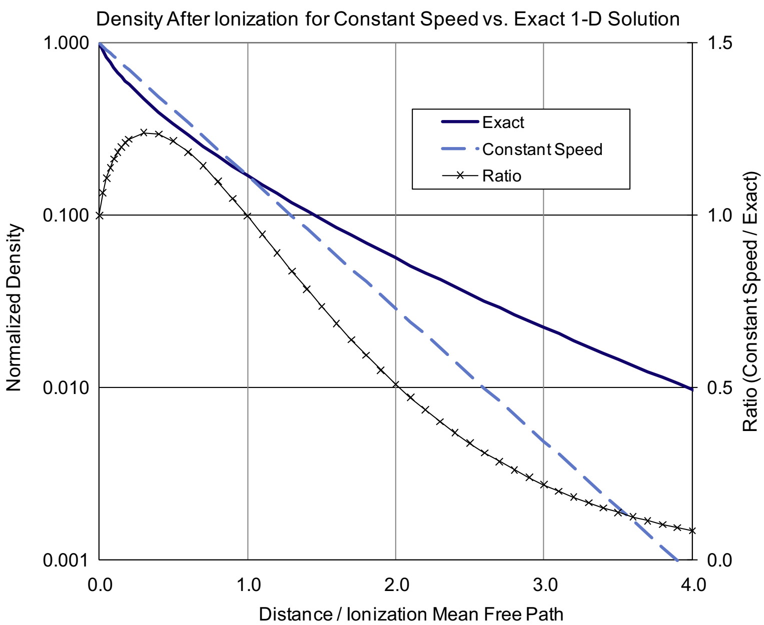

Fig. 10. Effect of assuming a constant-speed distribution on the density of a Maxwellian gas moving through a uniform ionizing medium.

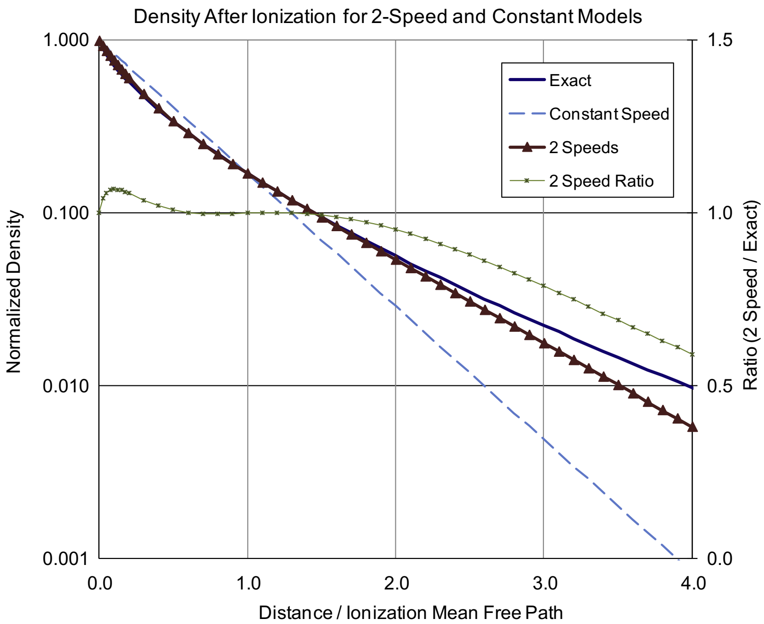

Fig. 11. Effect of using two speed bins to calculate the density of a Maxwellian gas moving through a uniform ionizing medium.

The improved accuracy is shown in Fig. 11. Having even two velocity bins reduces the maximum error by about a factor of four.

In the Hall thruster example above, the ionization region is very intense and very short. The effect of the constant-speed assumption moves ion generation less than one grid cell, and, since the algorithm is strictly mass conserving, is expected to have no impact on the total ion currents. The comparisons in Figs. 10 and 11 are shown on a logarithmic scale that accentuates the errors when most of the gas is ionized.

Another source of error is numerical diffusion associated with the first-order, upwind, algorithm used to advance the mass continuity equation. Since the velocities are calculated using geometrical shadowing, at large distances from the source there is little diffusion perpendicular to the flow direction.

5. Conclusion

A new algorithm has been developed for calculating neutral gas densities in computer simulations of plasma devices where the neutrals are in the free-molecule regime. The new algorithm has been implemented and used in a 2-D Hall thruster code that solves the full system of governing laws for the plasma. This new algorithm takes advantage of the fact that almost all neutral gas molecules in low density plasma devices have straight-line, constant-velocity, trajectories until they are either ionized, strike a wall, or leave the physical domain. The algorithm is based on the view factor approach used often for spacecraft contamination modeling, but includes time dependence and ionization. Compared with the PIC method commonly used in many plasma simulation codes, the new algorithm achieves “quiet” and smooth results. The first-order upwind differencing algorithm introduces some spatial diffusion. Like most numerical algorithms, higher accuracy can be achieved with finer resolution. The accuracy of the algorithm can be improved by finer spatial resolution of the emitting surfaces (this results in finer angular resolution of the velocity distribution function), and by dividing the Maxwellian speed distribution into two or more bins.

Acknowledgments

The authors thank Dr. Richard R. Hofer for providing the HPHall runs. The research described in this paper was carried out by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

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